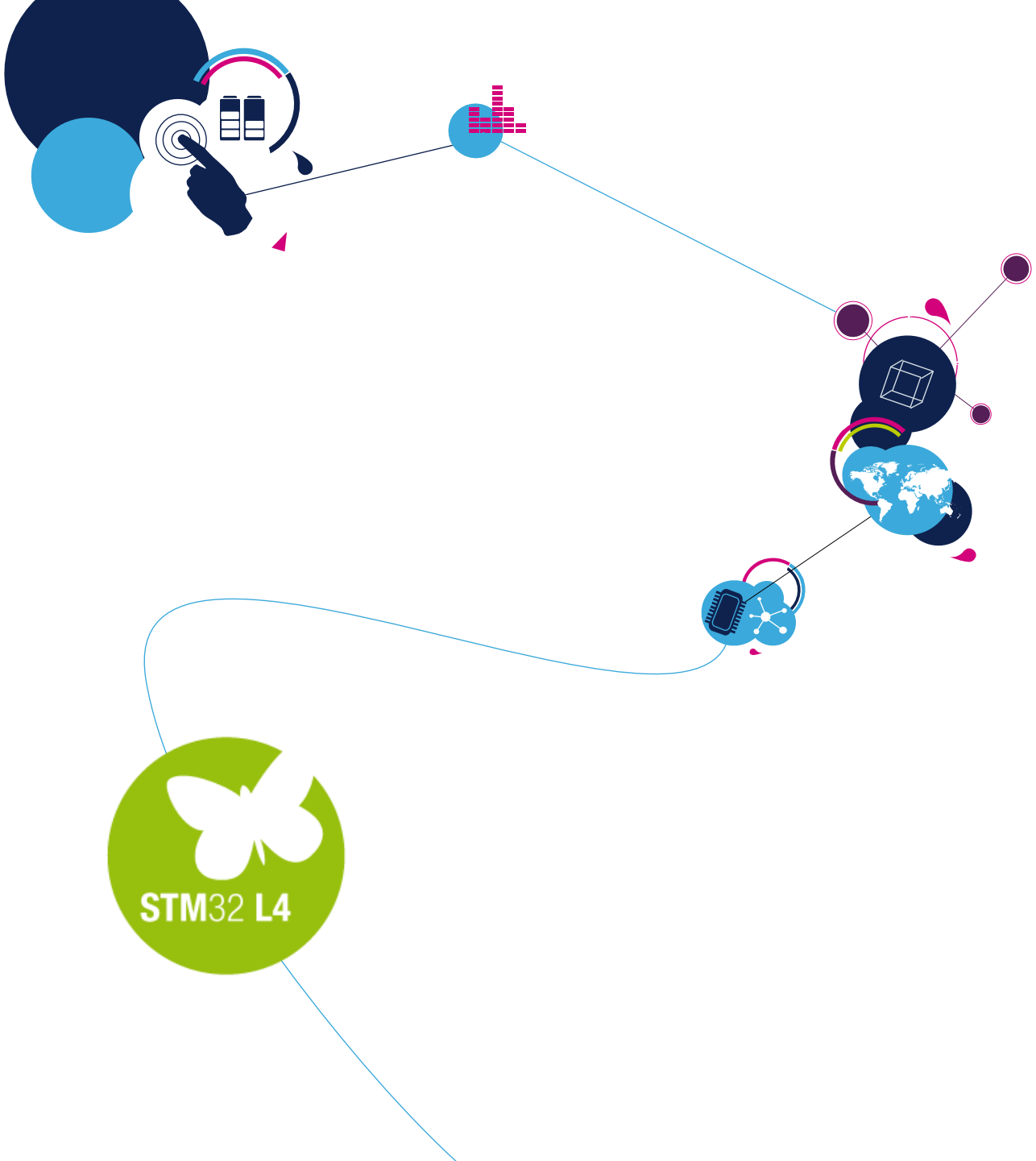
SW4STM32

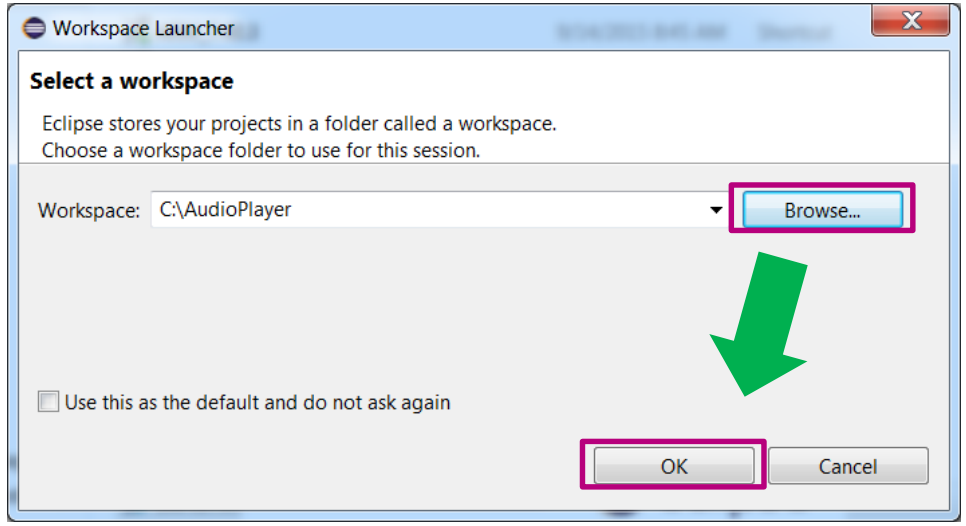
How to …



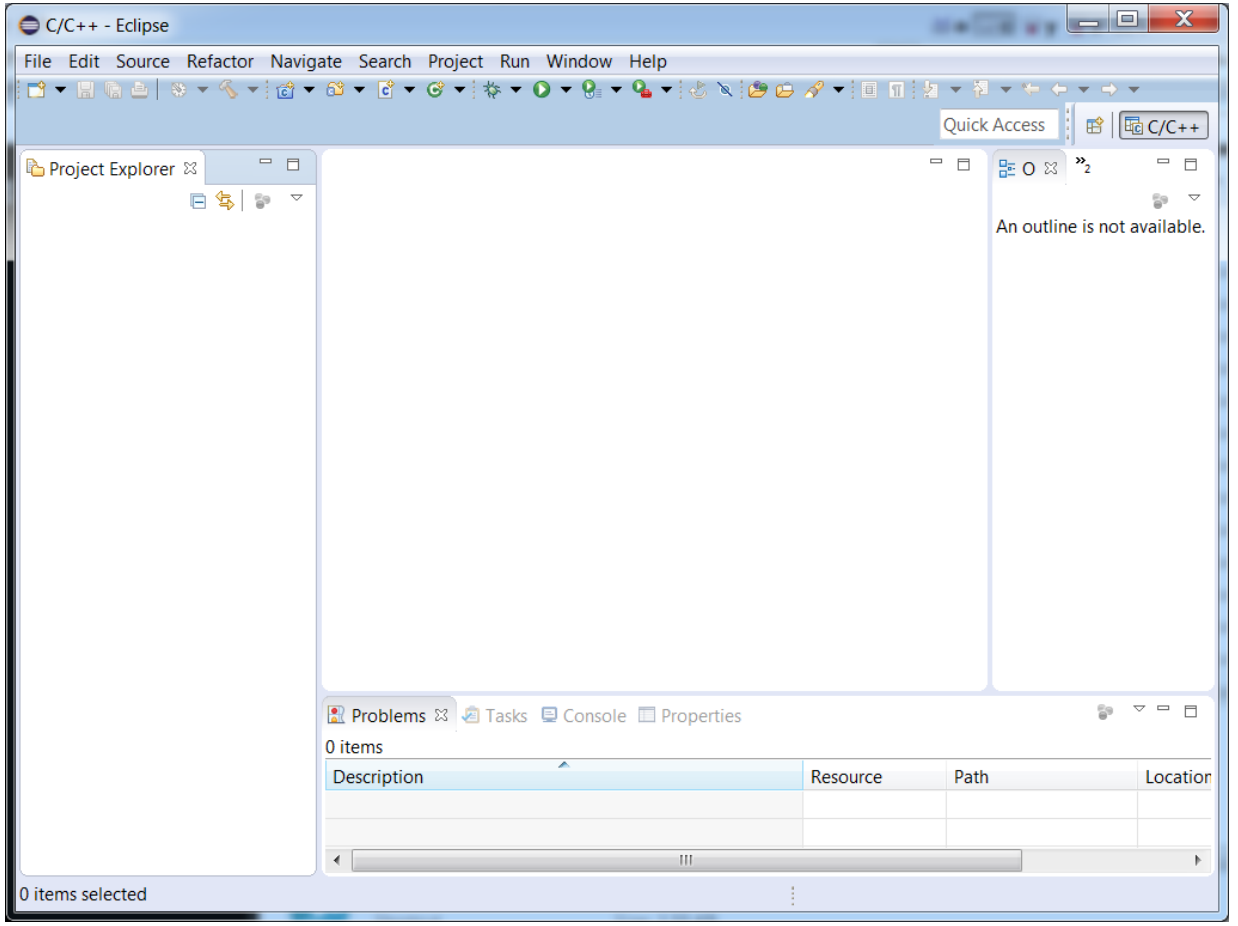
life.augmented

Create a new workspace

- Create a new workspace



- An empty workspace will be generated

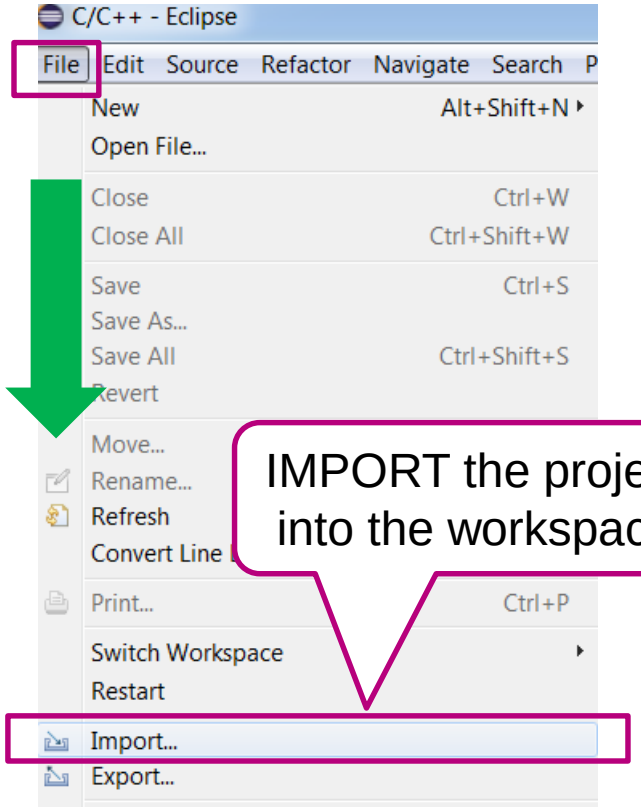


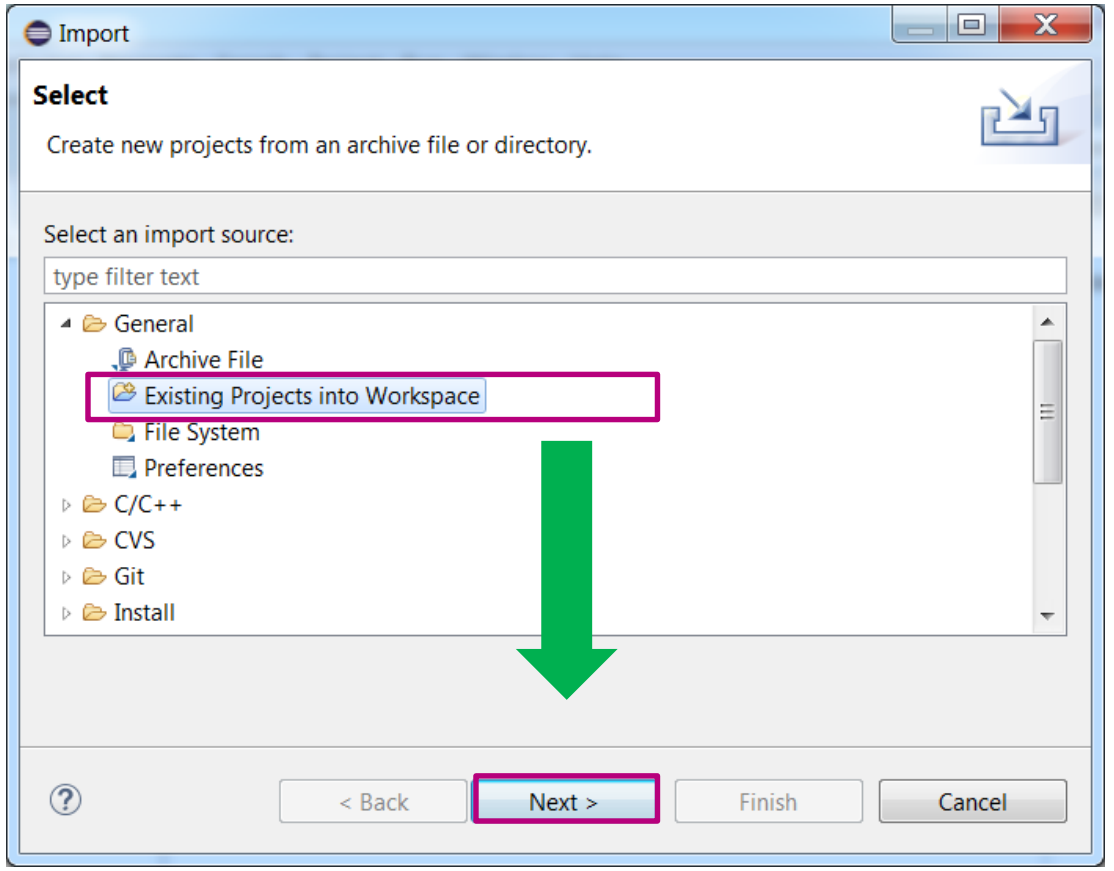
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Import the project into the workspace 1/3

SW4STM32

Unzip all SW4STM32 projects from **.\Hands-on\1_System_Operating_Modes.zip** archive



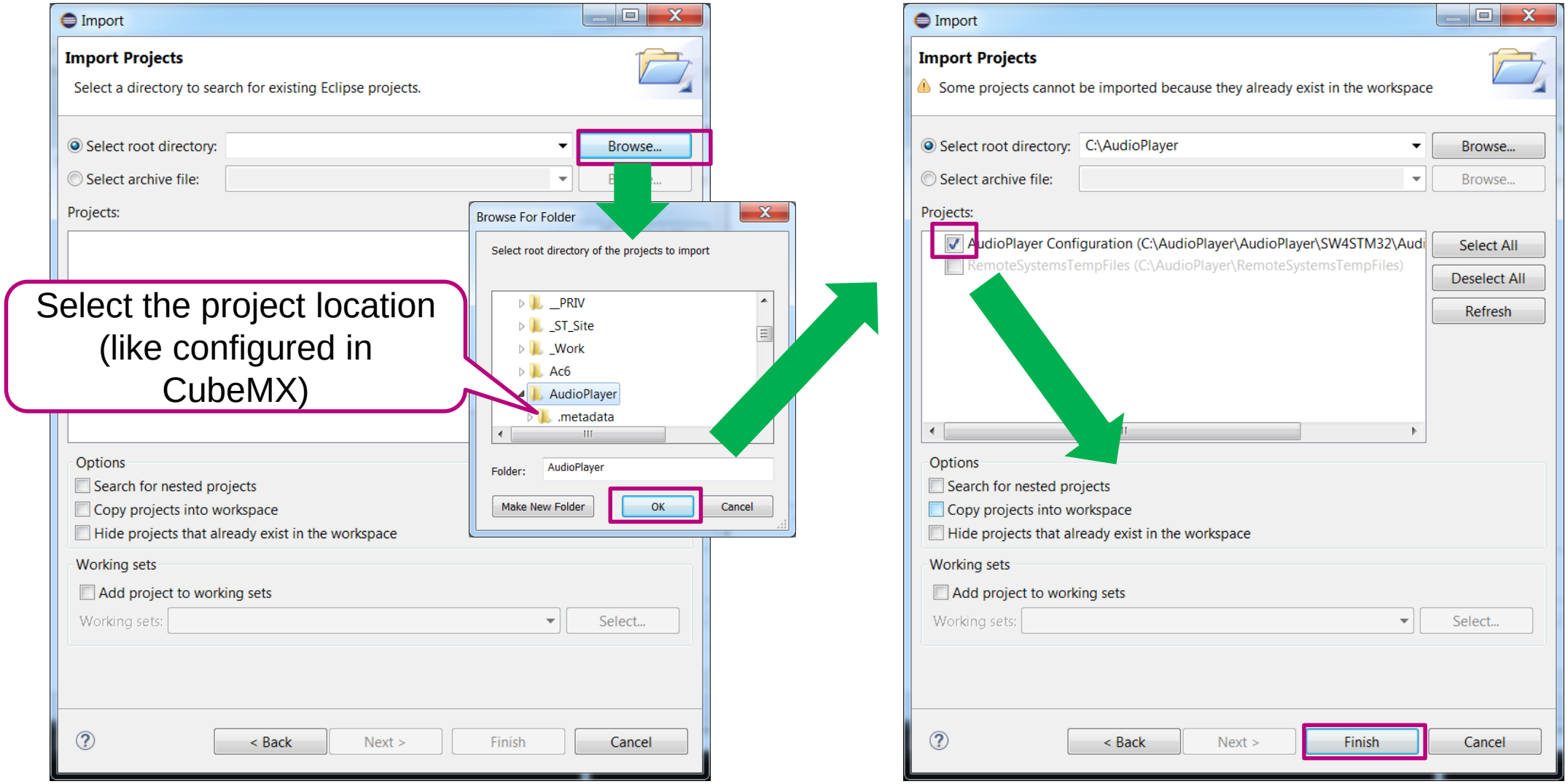


It is possible to import many project into one workspace

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Import the project into the workspace 2/3

SW4STM32



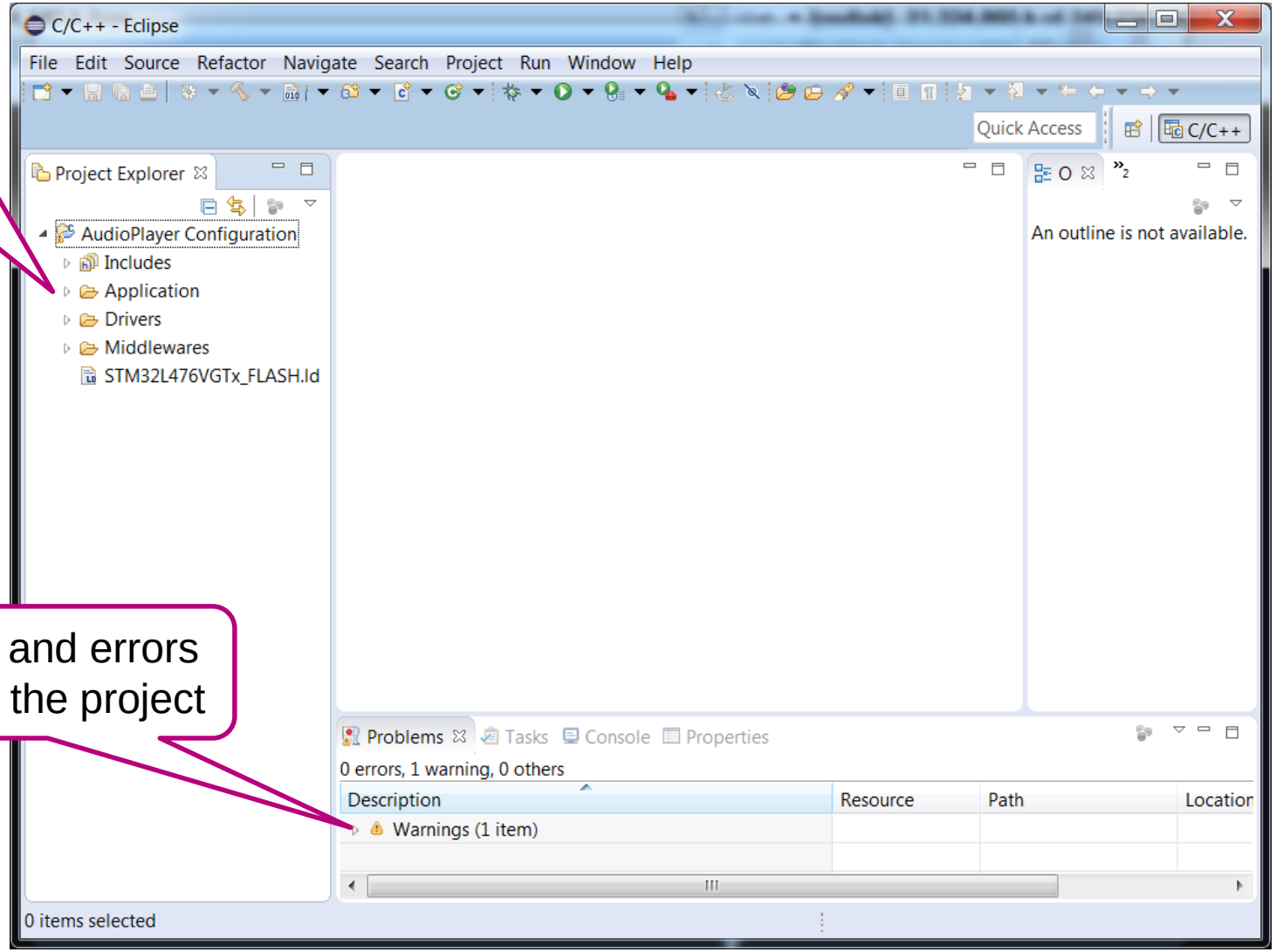
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Import the project into the workspace 3/3

SW4STM32

Once project is included into the workspace, its folder structure is visible in Project Explorer and the project is build automatically

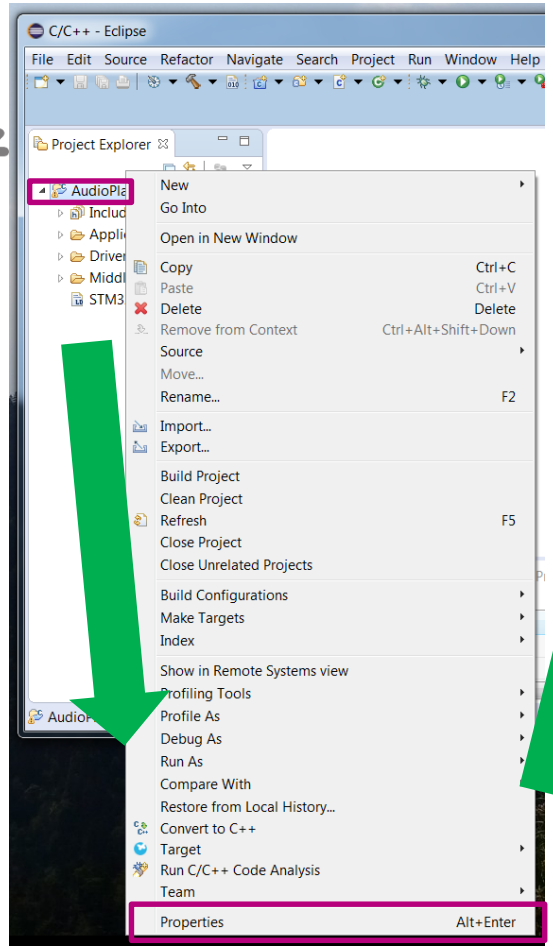
Warnings and errors after build the project

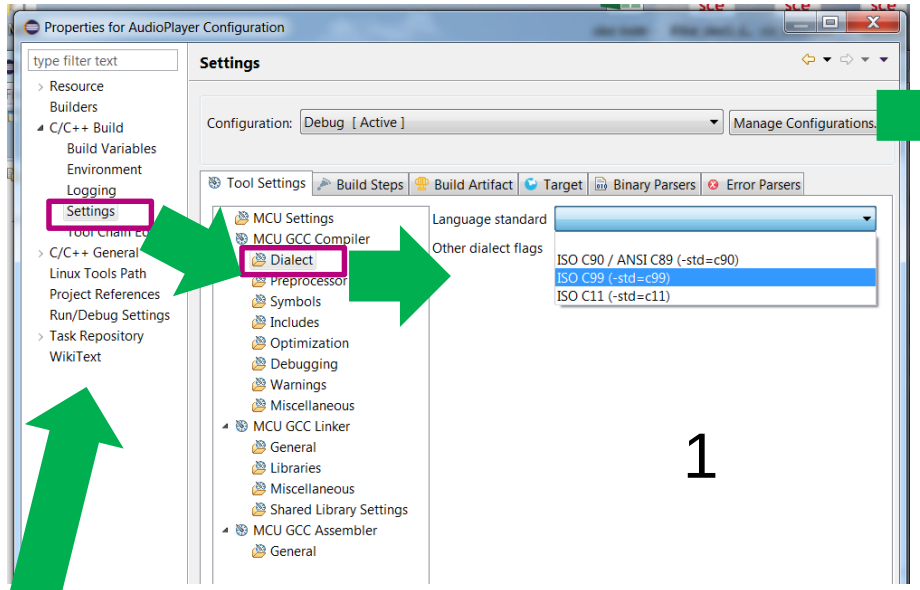


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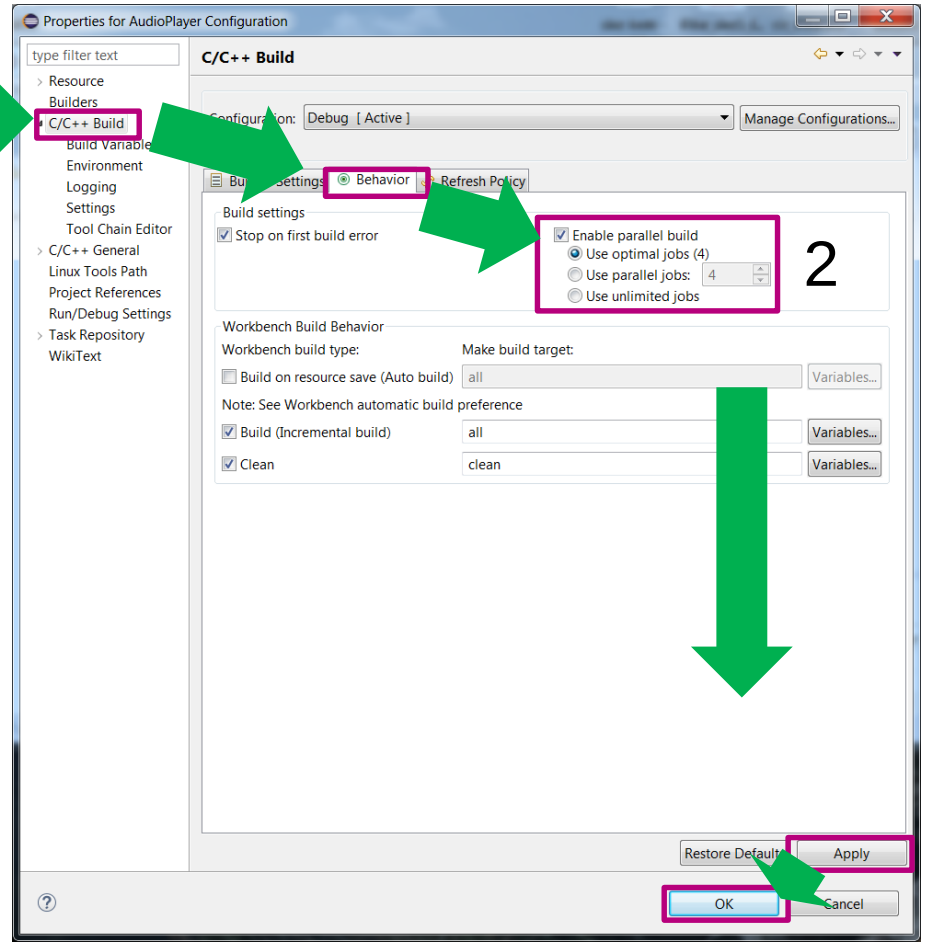
Project settings

SW4STM32





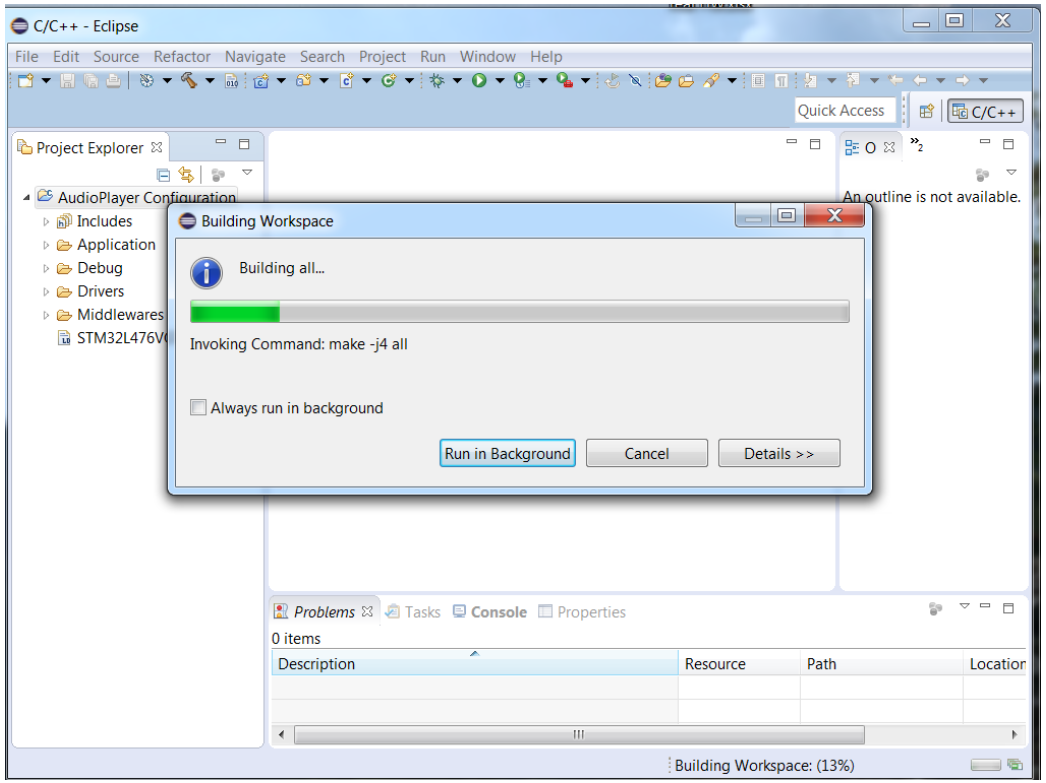
1. Configure C standard to C99 to avoid possible compilation errors
2. Enable parallel build to increase compilation speed



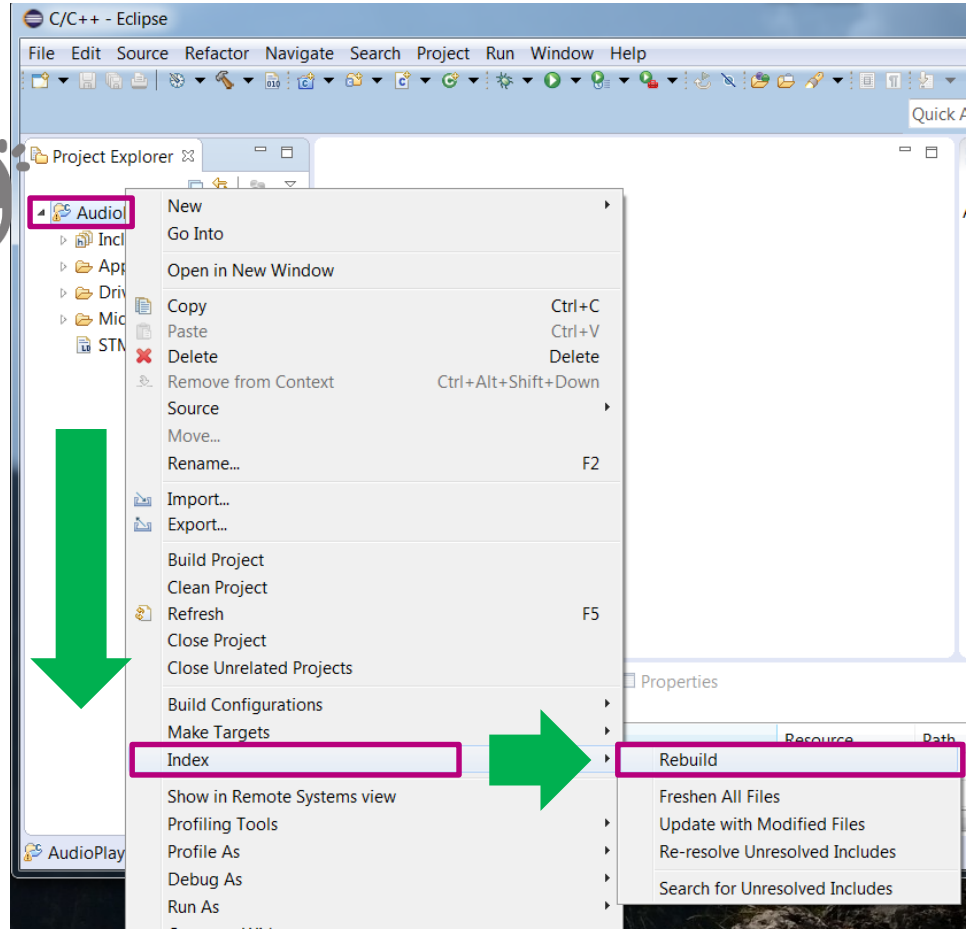
life.augmented

Build the project
SW4STM32

To build the project press Ctrl+B or select Make All button 



In case of multiple compilation errors, re-run Indexing for the project

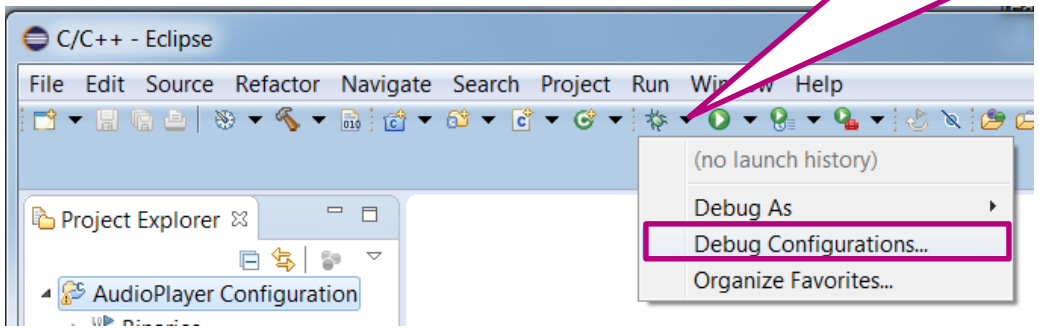


life.augmented

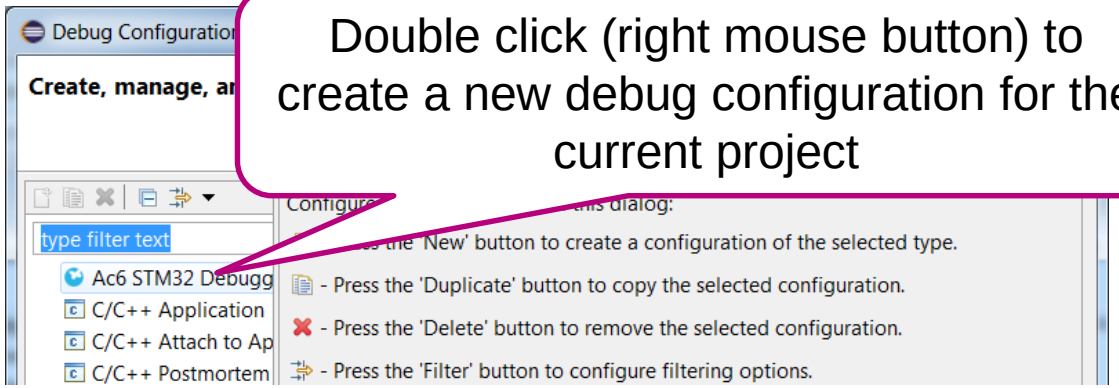
Configure the debug session (before first run)

SW4STM32

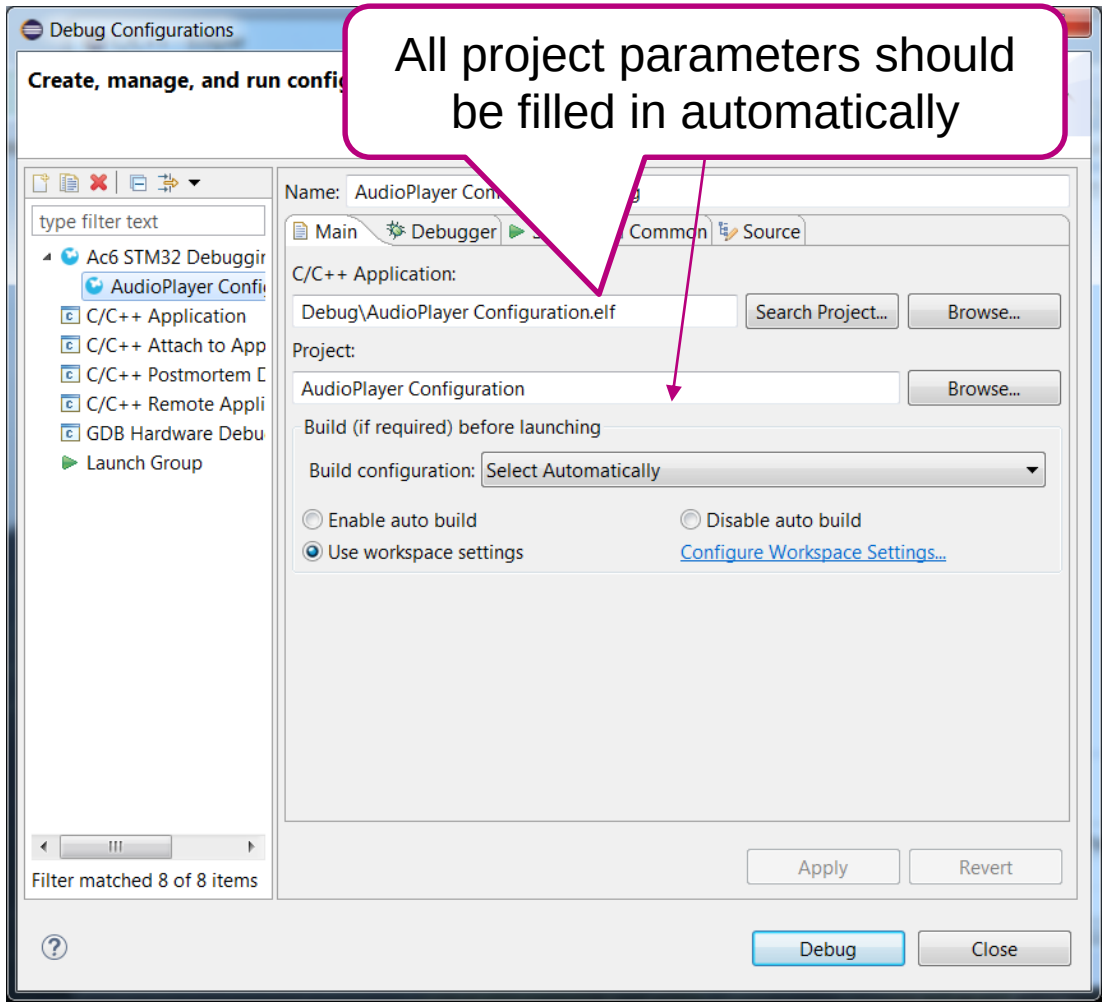
Before debug current project for the first time it is necessary to configure its debug session



Select Debug Configuration option



Double click (right mouse button) to create a new debug configuration for the current project



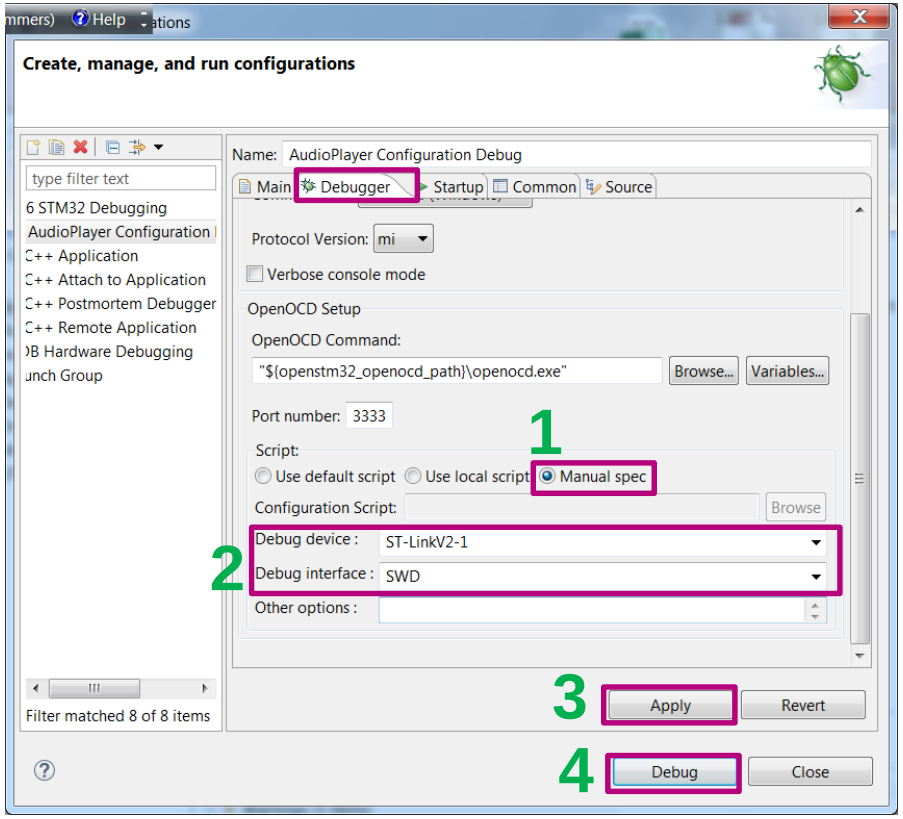
All project parameters should be filled in automatically

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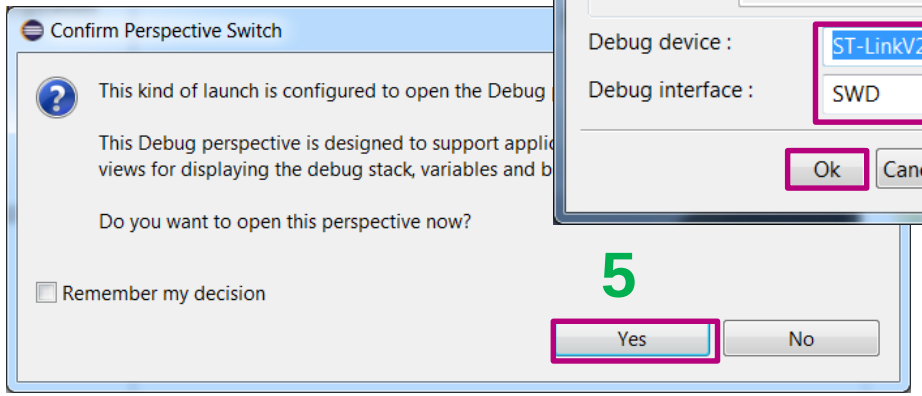
Run the debug session

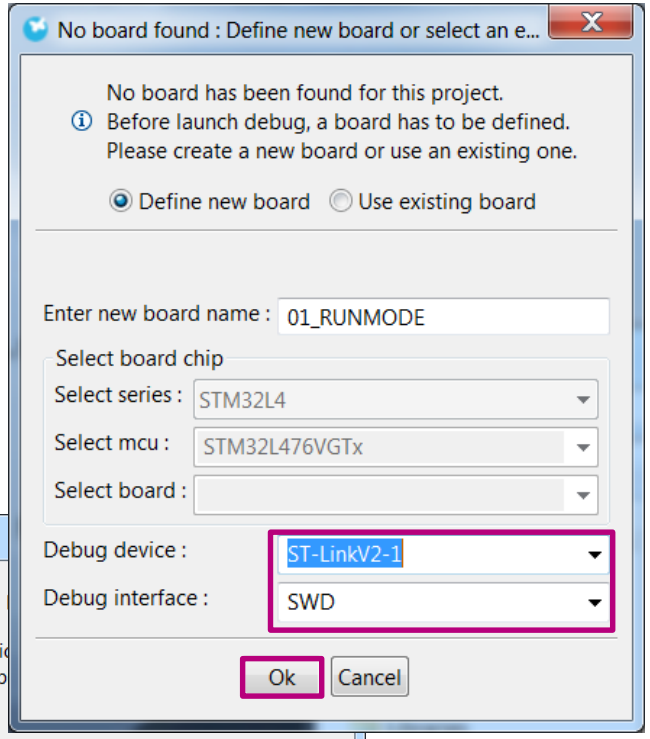
SW4STM32

- Connect discovery kit over miniUSB cable (ST-Link)
- Within Debugger tab select debug device (ST LinkV2-1) and debug interface (SWD)
- Select Apply and then Debug
- Debug perspective will be run (please select Yes on the information window)
- It will be enough to click “bug” button to enter debug session next time.



OR

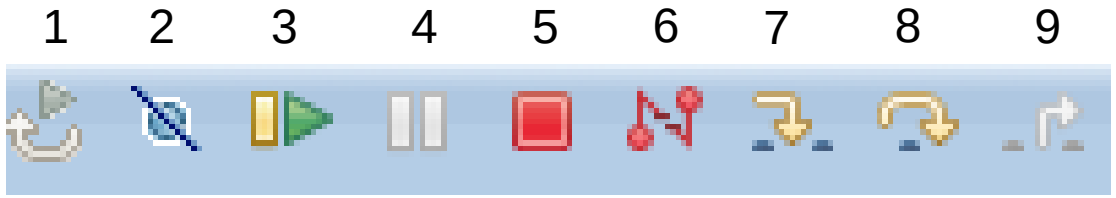




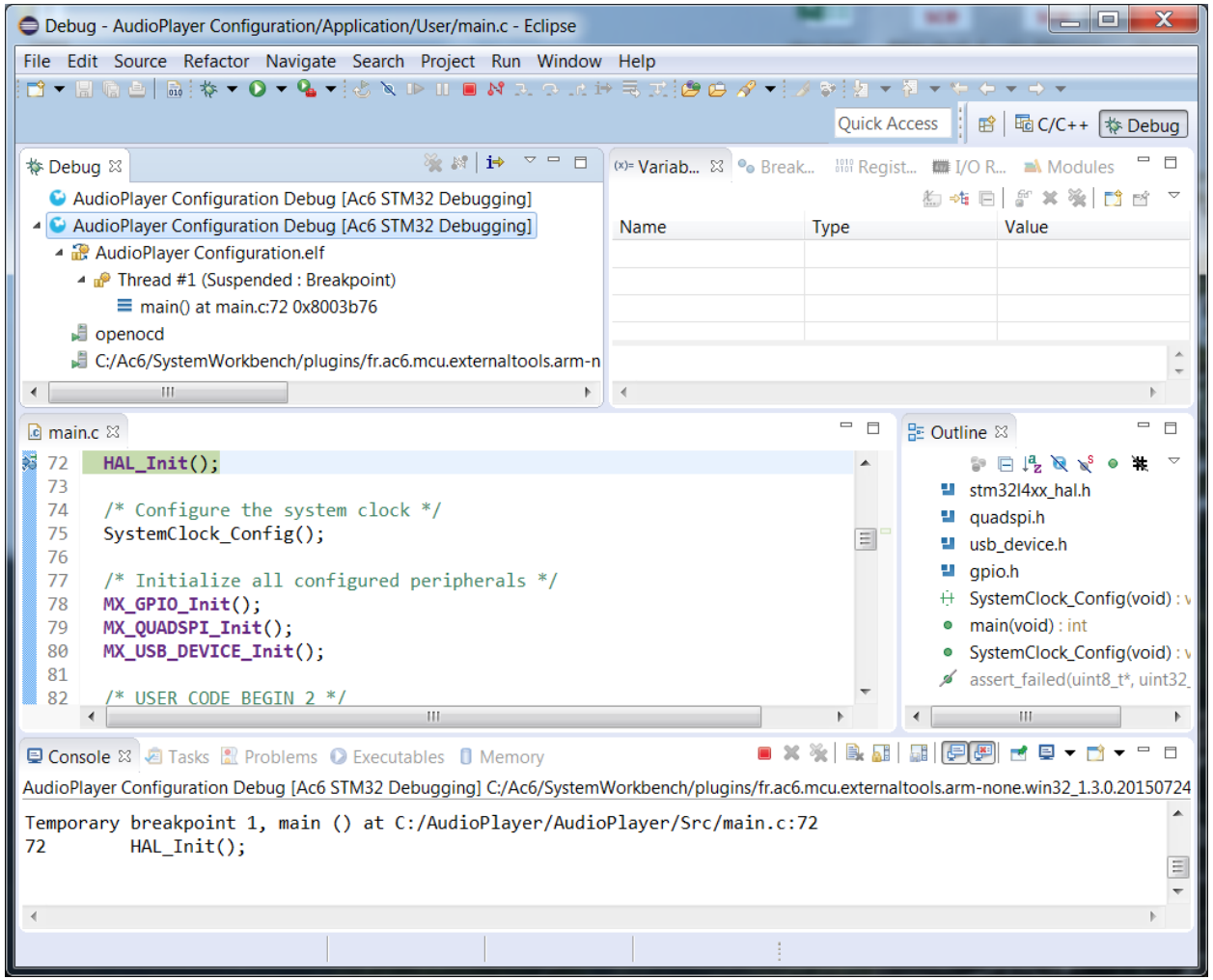
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Handling the debug session

SW4STM32



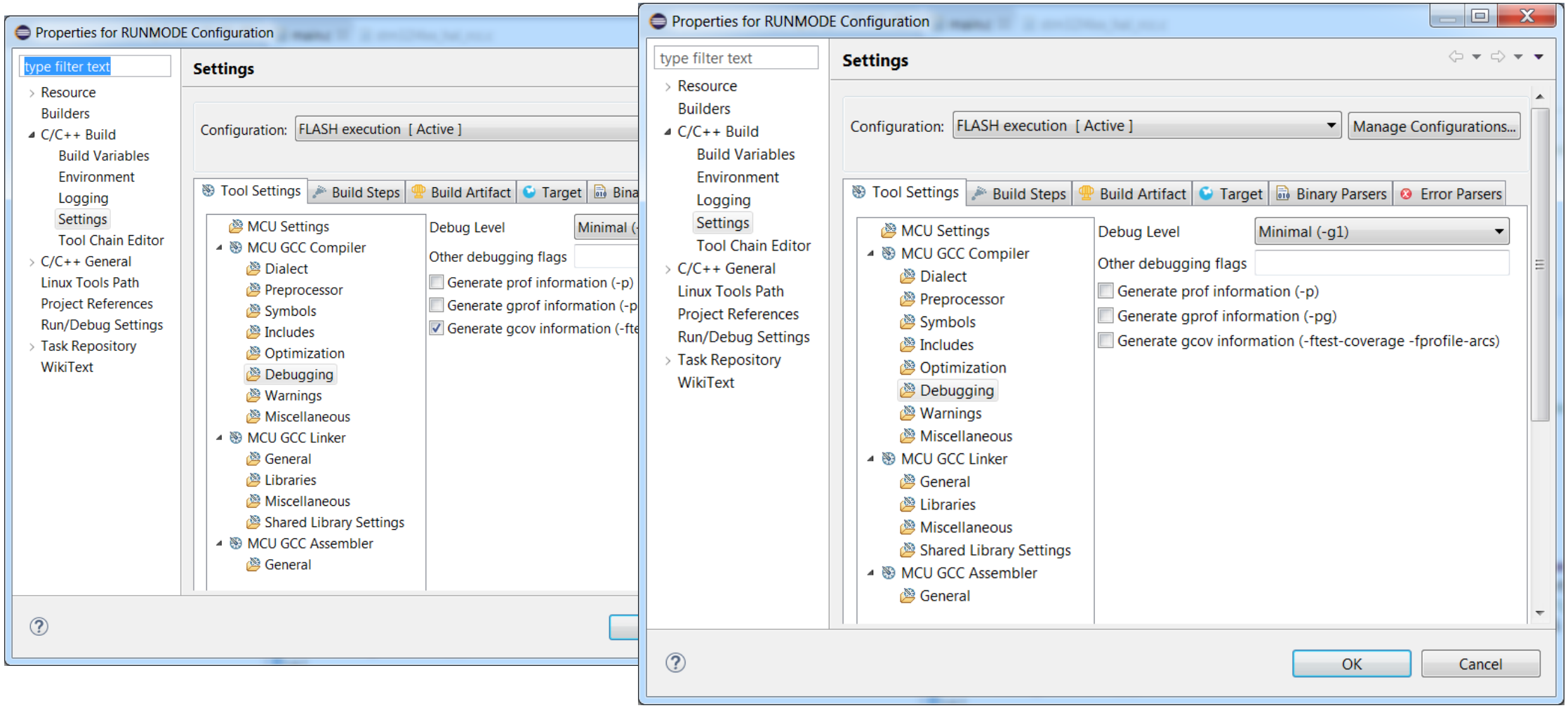
1. Reset
2. Skip all breakpoints
3. Run/resume
4. Suspend
5. Terminate debug session
6. Disconnect from the target
7. Step into
8. Step Over
9. Step Return



windows configuration in debug perspective

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Debug configuration

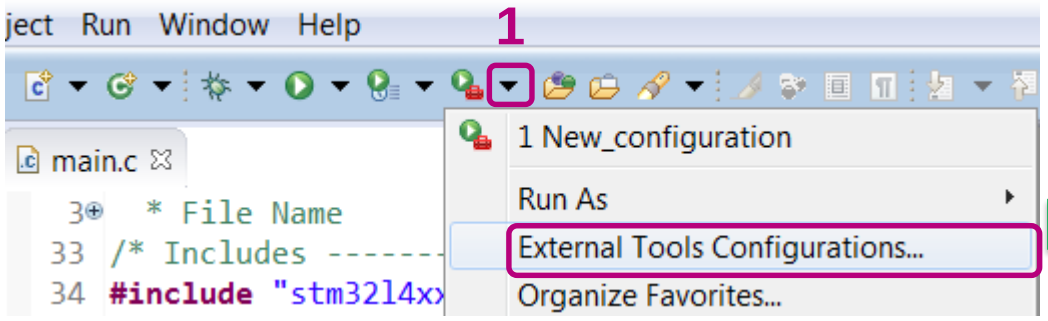


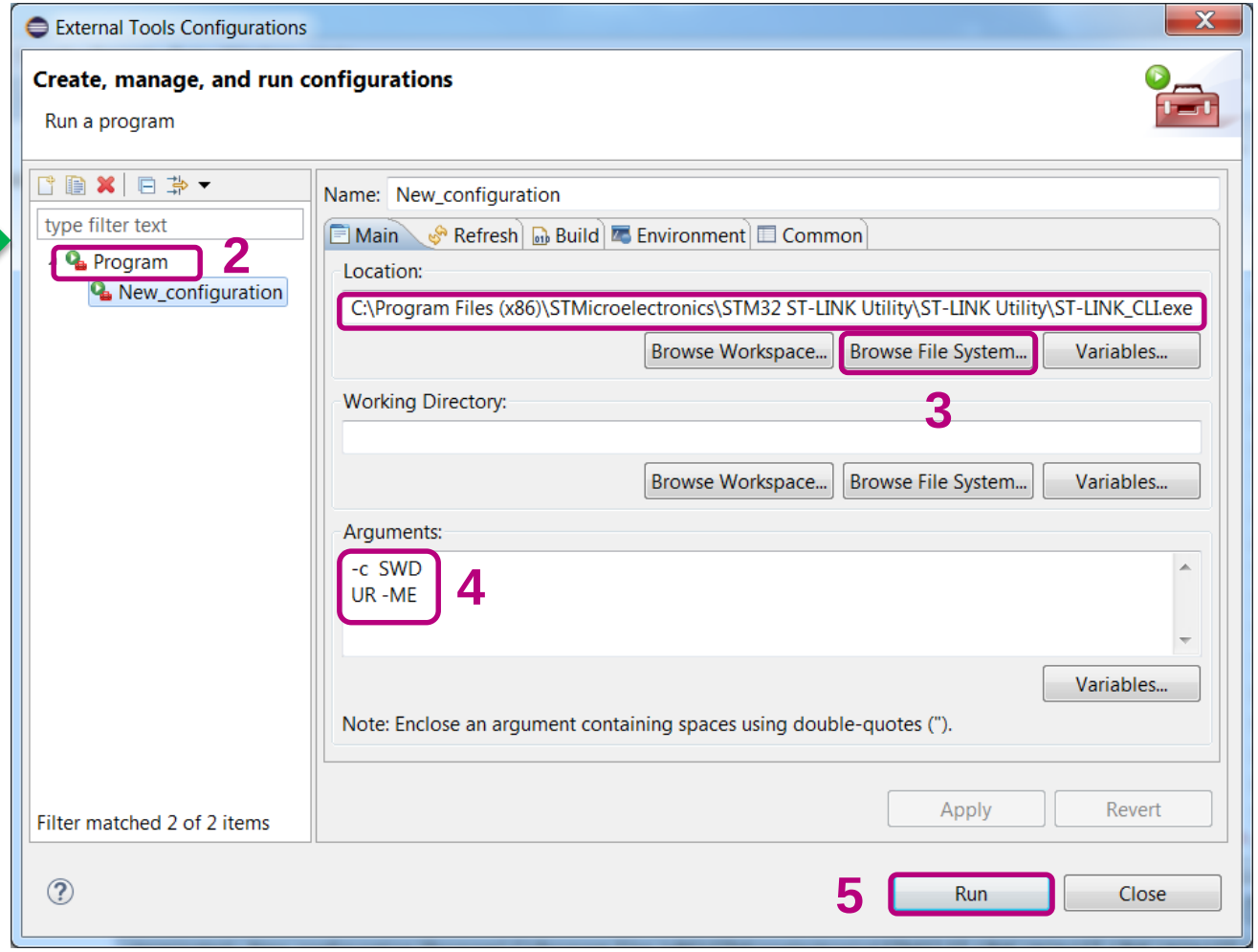
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There is no “connect under reset” option within OpenOCD

Run an external application

- How to erase Flash memory using command line STLink Utility application from SW4STM32





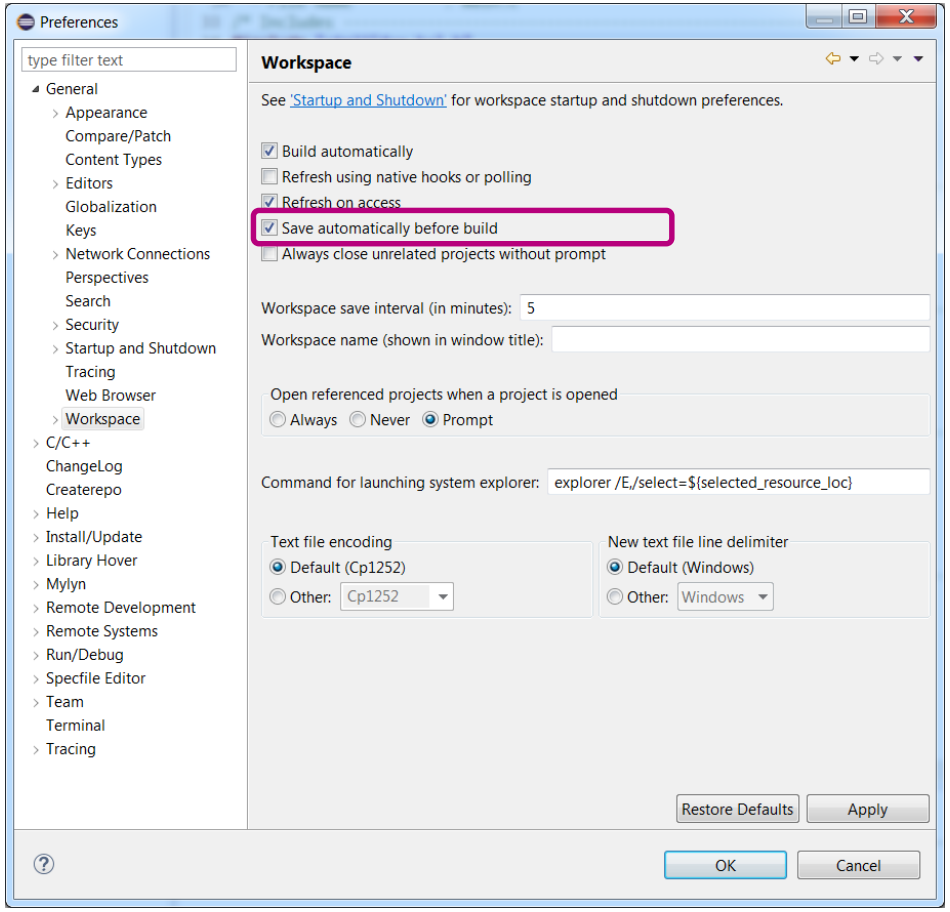
1. Select External Tools Configuration from Run configuration option
2. Create new configuration by double click on Program
3. Browse **ST-LINK_CLI.exe** within File System
4. Add –c SWD UR –ME arguments
   - connect by SWD [-c SWD]
   - with connect under reset option [UR]
   - and perform mass erase of the Flash memory [-ME]
5. Run the program (**ST-LINK_CLI.exe –c SWD UR –ME** will be executed)

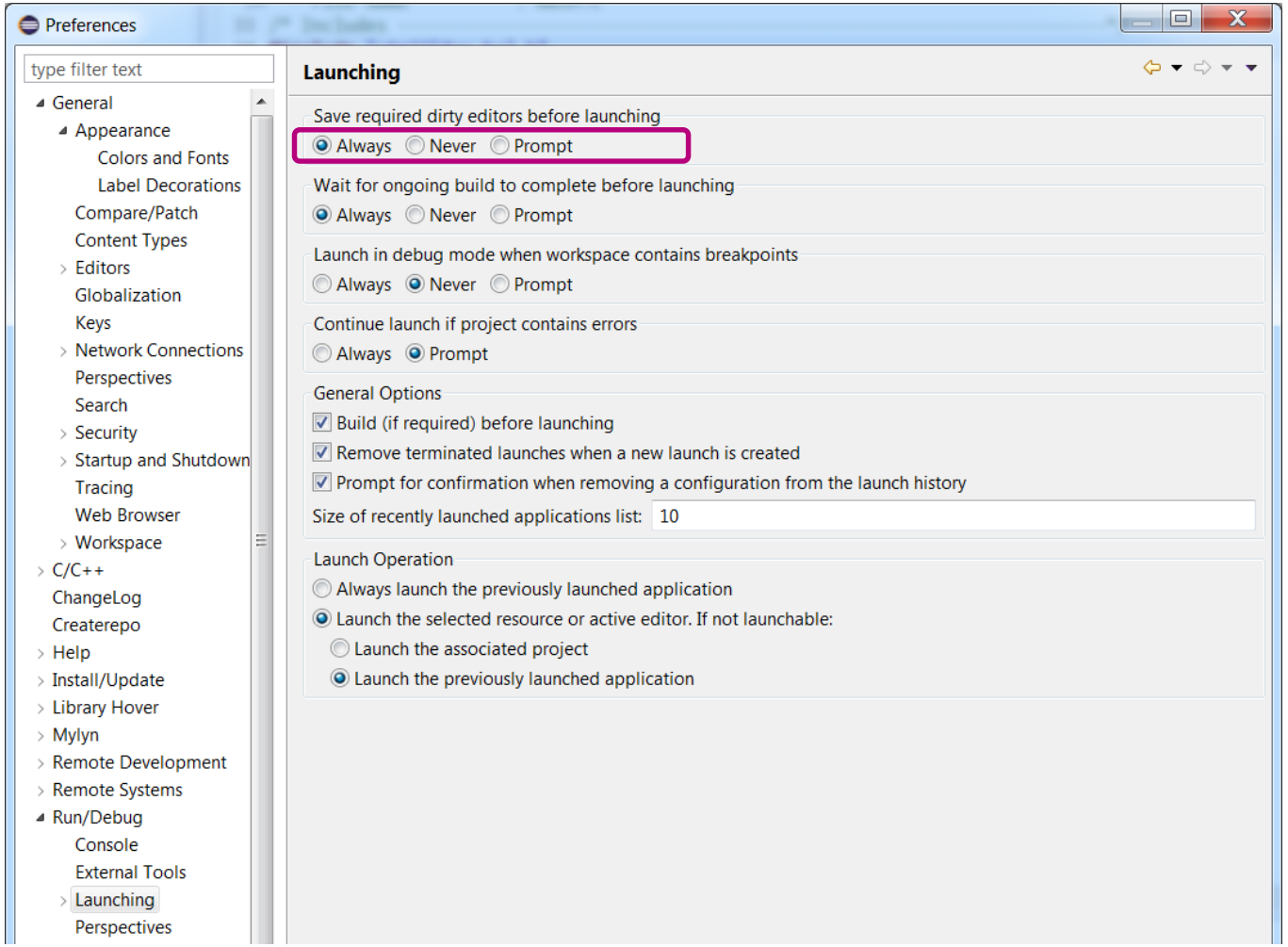
life.augmented

Useful configuration

autosave on build and launch

Editor settings are available in **Windows->Preferences** menu



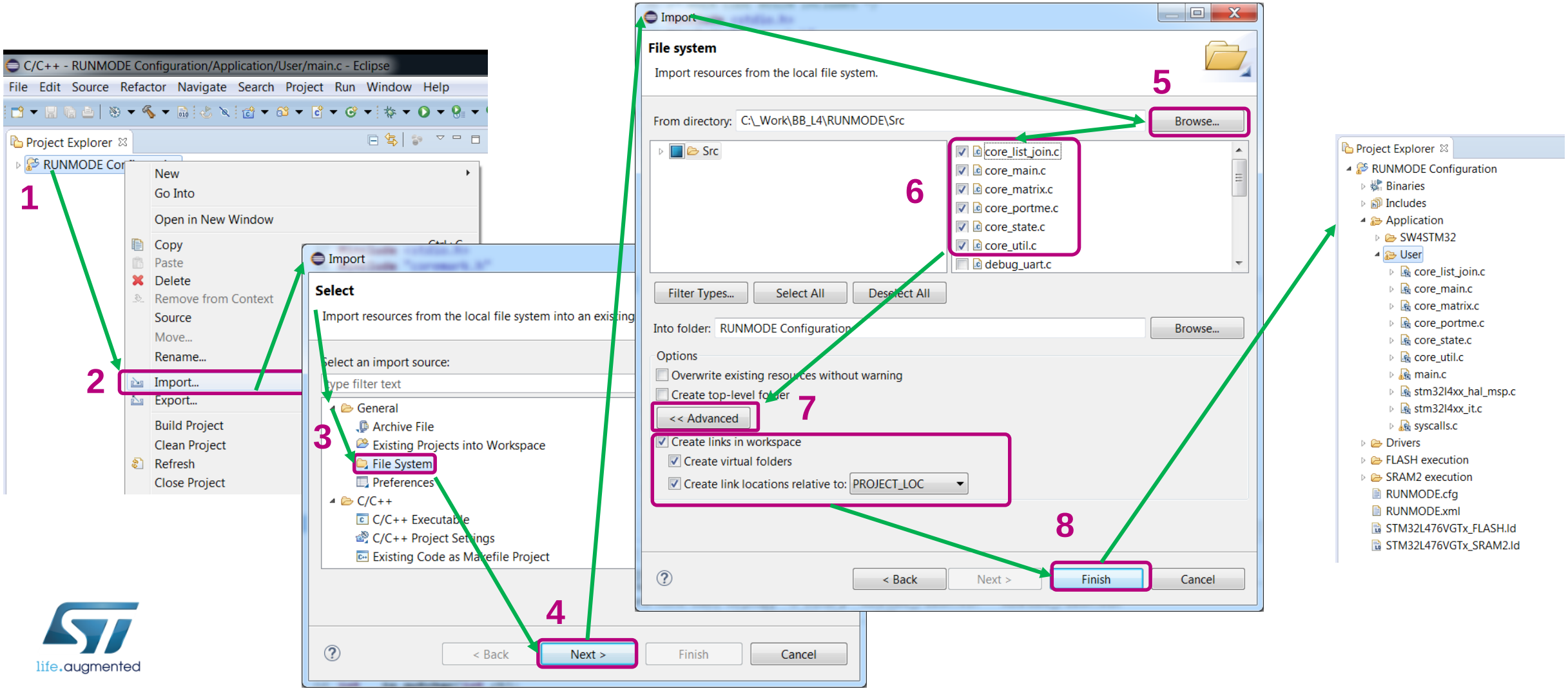


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Adding files to the project

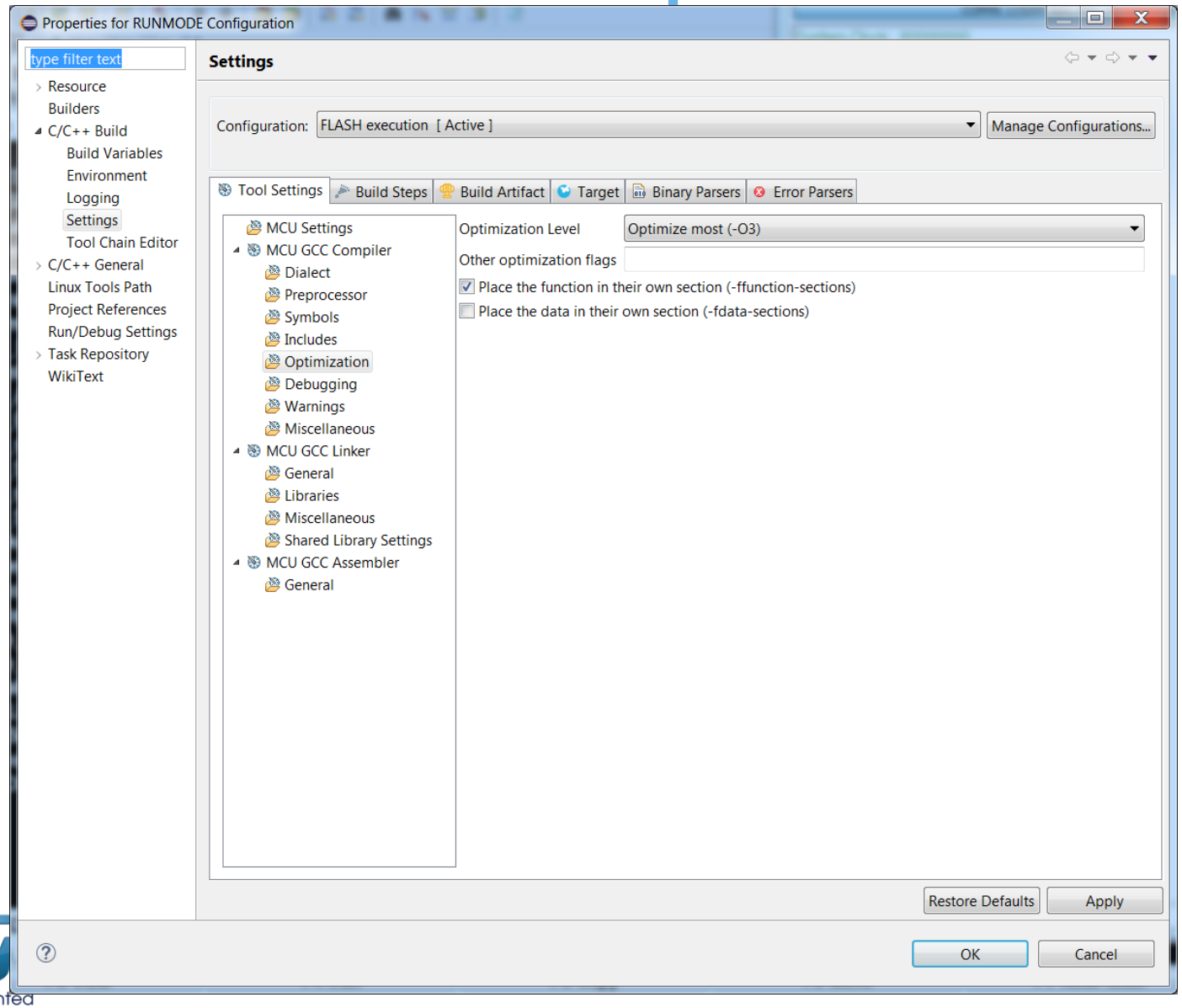
using link method (CubeMX style)

Select **IMPORT** menu command after right click on the files folder within the project



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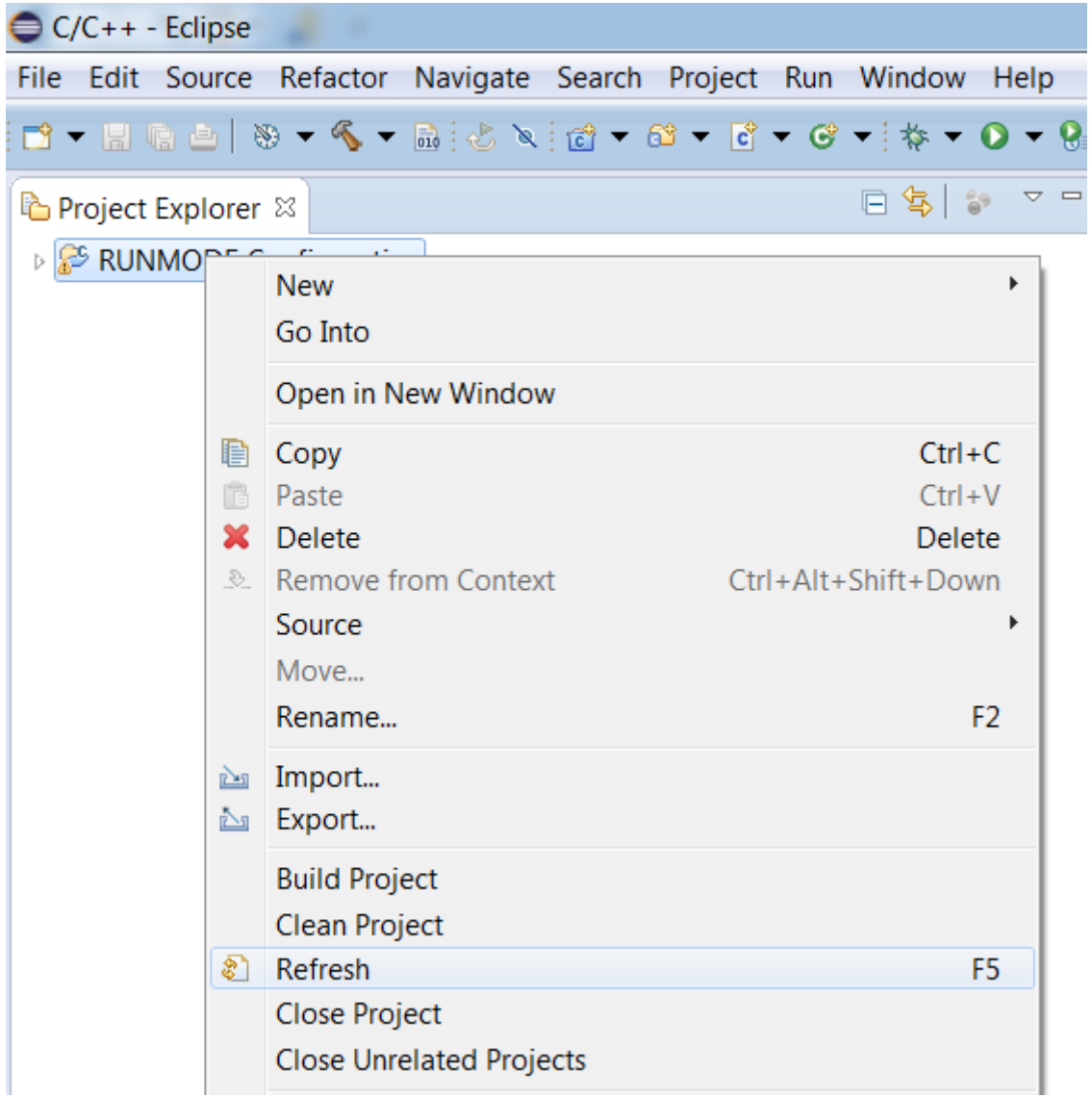
Optimization configuration



Without this option selection we have 2x more code

Project refresh after configuration change

- After performing any change on optimization/debug level, refresh project configuration (F5), then build it again



Enjoy!



www.st.com/stm32l4